

~~typed boring notes~~ -> my notes!



Types of Data



Session 1 on Descriptive Statistics.pdf

~ 13 Sep 2026 ~

- **Statistics** is a branch of mathematics that involves collecting, analysing, interpreting, and presenting **data**. It provides tools and methods to understand and make sense of large amounts of **data** and to draw conclusions and make decisions based on the **data**.
- In practice, **statistics** is used in a wide range of fields, such as business, economics, social sciences, medicine, and engineering. It is used to conduct research studies, analyse market trends, evaluate the effectiveness of treatments and interventions, and make forecasts and predictions.

Examples:

- Business - **Data Analysis** (Identifying customer behavior) and Demand Forecasting
- 1. Medical - Identify efficacy of new medicines (Clinical trials), Identifying risk factor for diseases (Epidemiology)
- 2. Government & Politics - Conducting surveys, Polling
- 3. Environmental Science - Climate research
- 4.

What is **Statistics**

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- **Descriptive statistics** deals with the collection, organization, analysis, interpretation, and presentation of **data**. It focuses on summarizing and describing the main features of a set of **data**, without making inferences or predictions about the larger **population**.
- Inferential **statistics** deals with making conclusions and predictions about a **population** based on a sample. It involves the use of

- probability theory to estimate the likelihood of
- certain events occurring, hypothesis testing to
- determine if a certain claim about a **population**
- supported by the **data** and regression analysis to
- examine the relationships between variables

Types of **Statistics**

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- **Population** refers to the entire group of individuals or objects that we are
- interested in studying. It is the complete set of observations that we want to make
- inferences about. For example, the **population** might be all the students in a
- particular school or all the cars in a particular city.
- A sample, on the other hand, is a subset of the **population** is a smaller group of
- individuals or objects that we select from the **population** to study. Samples are
- used to estimate characteristics of the **population**, such as the **mean** or the
- proportion with a certain attribute. For example, we might randomly select 100
- students.

Examples

- All cricket fans vs fans who were present in the stadium

1.

- All students vs who visit college for lectures

2.

- Things to be careful about while creating samples

Sample Size

1.

- Random

2.

- Representative

3.

Parameter Vs **Statistics**

- A parameter is a characteristic of a **population**, while a statistic is a characteristic
- of a sample. Parameters are generally unknown and are estimated using **statistics**.
- The goal of statistical inference is to use the information obtained from the sample

- to make inferences about the **population** parameters.
- **Population**'s Sample
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- **Session** on **Descriptive Statistics** Page 3
- Inferential **statistics** is a branch of **statistics** that deals with making inferences or predictions about a larger **population** based on a sample of **data**. It involves using statistical techniques to test hypotheses and draw conclusions from **data**. Some of the topics that come under inferential **statistics** are:
- Hypothesis testing: This involves testing a hypothesis about a **population** parameter based on a sample of **data**. For example, testing whether the **mean** height of a **population** is different from a given value.
- 1.
- Confidence intervals: This involves estimating the range of values that a **population** parameter could take based on a sample of **data**. For example, estimating the **population mean** height within a given confidence level.
- 2.
- Analysis of variance (ANOVA): This involves comparing means across multiple groups to determine if there are any significant differences. For example, comparing the **mean** height of individuals from different regions.
- 3.
- Regression analysis: This involves modelling the relationship between a dependent variable and one or more independent variables. For example, predicting the sales of a product based on advertising expenditure.
- 4.
- Chi-square tests: This involves testing the independence or association between two categorical variables. For example, testing whether gender and occupation are independent variables.
- 5.
- Sampling techniques: This involves ensuring that the sample of **data** is representative of the **population**. For example, using random sampling to select individuals from a **population**.
- 6.
- Bayesian **statistics**: This is an alternative approach to statistical inference that involves

- updating beliefs about the probability of an event based on new evidence. For example,
- updating the probability of a disease given a positive test result.
- 7.

- Why ML is closely associated with **statistics**?

- Inferential **Statistics**

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- A measure of central tendency is a statistical measure that represents a typical or
- central value for a dataset. It provides a summary of the **data** by identifying a
- single value that is most representative of the dataset as a whole.

- Measure of Central Tendency

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- **Mean** The **mean** is the sum of all values in the dataset divided by the number of
- values.

- 1. **Mean**

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- Median: The median is the middle value in the dataset when the **data** is arranged
- in order.

- 2. Median

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- Mode: The mode is the value that appears most frequently in the dataset.

- 3. Mode

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- Weighted **Mean**: The weighted **mean** is the sum of the products of each value and its weight, divided by the sum of the weights. It is used to calculate a **mean** when the values in the dataset have different importance or frequency.

- 4. Weighted **Mean**

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- A trimmed **mean** is calculated by removing a certain percentage of the smallest and largest values from the dataset and then taking the **mean** of the remaining values. The percentage of values removed is called the trimming percentage.

- 5. Trimmed **Mean**

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- A measure of dispersion is a statistical measure that describes the spread or variability of a dataset. It provides information about how the **data** is distributed around the central tendency (**mean**, median or mode) of the dataset.

- Measure of Dispersion

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- Range: The range is the difference between the maximum and minimum values in the dataset. It is a simple measure of dispersion that is easy to calculate but can be affected by outliers.

- 1. Range

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Variance: The variance is the average of the squared differences between each

data point and the mean. It measures the average distance of each data point

from the mean and is useful in comparing the dispersion of datasets with different

means. *

- $\sum (X - \text{mean})^2$
- 3-3
- 2-3
- 1-3
- 5-3
- 4-3

Mean Absolute Deviation *

2. Variance *

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- **Standard Deviation** The **standard deviation** is the square root of the variance. It is
- a widely **used** measure of dispersion that is useful in describing the shape of a
- distribution.
- 3. **Standard Deviation**
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- **Coefficient of Variation (CV)**: The CV is the ratio of the **standard deviation** to the
- mean expressed as a percentage. It is **used** to compare the variability of datasets

- with different means and is commonly **used** in fields such as biology, chemistry,
- and engineering.
- The **coefficient of variation** (CV) is a statistical measure that expresses the amount
- of variability in a dataset relative to the mean. It is a dimensionless quantity that is
- expressed as a percentage.
- The formula for calculating the **coefficient of variation** is:

$$CV = (\text{standard deviation} / \text{mean}) \times 100\% *$$

- 4. **Coefficient of Variation**
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Graphs for Univariate Analysis *

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- A **frequency distribution table** is a **table** that summarizes the number of times (or
- **frequency**) that each value occurs in a **dataset**.
- Let's say we have a survey of 200 people and we ask them about their favourite
- type of vacation, which could be one of six categories: Beach, City, Adventure,
- Nature, Cruise, or Other
- Relative **frequency** is the proportion or percentage of a **category** in a **dataset** or
- **sample**. It is calculated by dividing the **frequency** of a **category** by the total number
- of observations in the **dataset** or **sample**.
- Cumulative **frequency** is the running total of frequencies of a variable or **category**
- in a **dataset** or **sample**. It is calculated by adding up the frequencies of the current
- **category** and all previous categories in the **dataset** or **sample**.
- 1. Categorical - **Frequency Distribution Table** & Cumulative **Frequency**
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- Shapes of Histogram

2. Numerical - Frequency Distribution Table & Histogram *

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- Session 1 on Descriptive Statistics Page 20

Graphs for Bivariate Analysis *

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- Contingency Table/Crosstab
- A **contingency table**, also known as a cross-tabulation or crosstab, is a type of **table**
- used in **statistics** to summarize the relationship between two **categorical** variables.
- A **contingency table** displays the **frequencies** or relative **frequencies** of the
- observed values of the two variables, organized into rows and columns.
- 1. **Categorical Categorical**
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Scatter Plot *

2. Numerical - Numerical *

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3. Categorical - Numerical *

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